How to design your own BNC Female calibration Kit with verified performance

Preface:

The BNC connector is very much used within the radio amateur community, and many other places, but calibration kits with verified performance and defined data for use with a Vector Network Analyzer, such as the famous VNWA developed by Tom Baier and sold by SDR-Kits, and the N2PK designed by Paul Kiciak, has only been available from professional sources and so expensive that it is out of reach for non professionals.
There exist many attempts to create a homemade BNC Female calibration standard, as well for a male calibration standard, but all suffer more or less from defined data such as delays, impedances and fringe/stray components.
Thus article demonstrates:
- how to design such a Female calibration kit very cheaply
- how to derive the calibration data and
- how to test the performance using the CT-Check method described long ago by Rohde & Schwarz.

The method described here can very well be copied to other sort of Connectors, such as the N Type which suffer from similar availability problem.

Some few initial statements:

We are not dealing with "rocket science" and a pragmatic approach has been taken to reach my favorite goal for a solution to a problem being "good enough is best". I like perfection but taking into account a delay of 1 ps is correspond to 0.2mm for a BNC adaptor with FTFE, and that disconnect/connect a BNC adaptor can lead to + - 1ps variation then it is overkill to aim for many decimal points for delays. Many decimal points for delays "does not hurt or help" so to speak. However you will see figures in this document with two decimal points, as that is the limit for the safe visual resolution limit for the VNWA, and thus kept recorded until additions of several delays is made. A delay of 0.01ps correspond to 0.002mm, that is not much of a distance.

Let us get started:

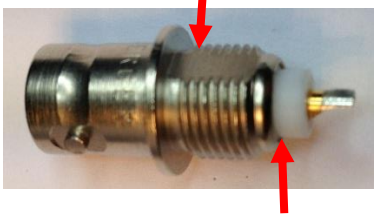
<u>Conditions are:</u>

1. That you must own a VNWA2 or VNWA3 and the associated software. However N2PK VNA owner has also a possibility by using the VNWA software controlling the N2PK VNA.
Although the following measurement is done for the entire frequency range up to 1.3GHz, I recommend to limit the frequency range to 500MHz, as you then are not disturbed by the odd events in the frequency range from 800MHz and upwards. When finished with the determination of all the parameters it is quite easy to check the performance up to 1.3Gz with a T-Check (se later) and eventual do some fine tuning if it at all helps. Likewise the 10dB adaptor shown in the setup picture can be omitted, as you might not have such a device. It is merely used to improve the T-Check in the frequency range from 700MHz upwards as discussed later. (A 20 dB attenuator is even better in the 1 to 1.3GHz range. Still the results without any attenuator is fully acceptable with a T-Check up to 1.3GHz .

2. You must get hold of minimum 5 of below shown Female BNC Flange connectors
- One is used for mounting 2x100ohm 0.1% SMD 0803 resistors with full length and with center pin grinded flush with end of the metal part.
- One is used to produce a "0 ps" delay Short Reference Standard as detailed for reference purposes
- One is used for producing a Short of exactly same length as the Load Standard, to be able to find the accurate delay for the Open and Load Standard, based on comparison with the 0 pS Short Reference Standard.
- One is used for producing an Open Standard where we also can calculate or simulate the delay based on mechanical measurements with a caliper or better instrumentation if available.
- One is used for producing an alternative Open standard, which is soldered to the back of the 0 ps Short Standard else very hard to fit the 0 ps Short Standard on a BNC male connector.

Do use BNC adaptor from well know manufacturers such as Huber Suhner, Rosenberger, Amphenol Connex, and several other well-known companies. It might be tempting to buy form Jo's around the corner but often cheap stuff is not with PTFE but white plastic and the there is no chance to use my information about delay as in the end of the document is a work plan if you just want to clone my design with adaptor from a good source and be able to calculate correct delays even if the adaptor length differ a few mm.
- Also purchase two female female adaptors

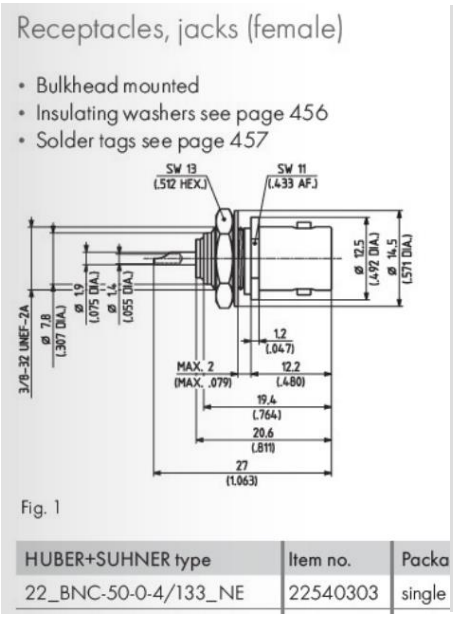
Cut the end off with a metal saw, to about 1 mm above the flange, and grind/mill it flat to flange surface
Mount it on a Male BNC while grinding. Make 3 of these and remove center pin and PTFE bushing.
The shown middle bushing is the residual from cutting of the part to be used for the 0 ps delay Short.



Cut off with sharp knife this disk for producing Load and a test Short
For the test short keep the center conductor and for Load grind it down to the end of metal

It will be demonstrated that when we have an accurate 0 ps Short Reference Standard and an accurate 50 ohm load with known delay (based on measurement of the Short Standard and compare to the 0ps Reference Short) we can derive the remaining calibration data and by use of a T-Check measurement verify the quality of the calibration based on our home made Load, 0 ps Short and the Open Standards.
Further we need a Female-Female BNC Adaptor and a Female-Male-Female BNC T-Adaptor or even better a homemade T-Check Adaptor based on a Female-Female BNC Thru adaptor, with a 2 mm slot in the center and where two 100 ohm 1% (or 0.1%) SMD 1206 Resistors are soldered between center conductor and "ground". Thus the center is loaded with 50 ohm as required for the T-Check. See pictures below. Use the same type of Thru adaptor for Thru calibration and T Check adaptor (same delay) as we then can derive the accurate delay of the Thru adaptor required for the Calibration Settings. Open and Thru adaptor delays will be found during a T-Check procedure.

Thru Adaptor and T-Check adaptor
2 x 100 ohm 0.1% SMD 1206 resistors in the center as accurate as possible

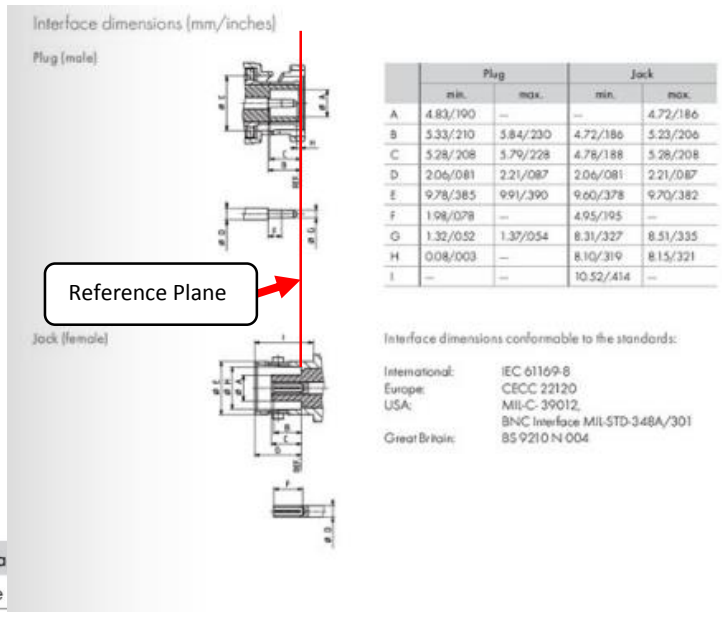


Receptacles, jacks (female)

- Bulkhead mounted
- Insulating washers see page 456
- Solder tags see page 457

Fig. 1

HUBER+SUHNER type	Item no.	Packa
22_BNC-50-0-4/133_NE	22540303	single



Interface dimensions (mm/inches)

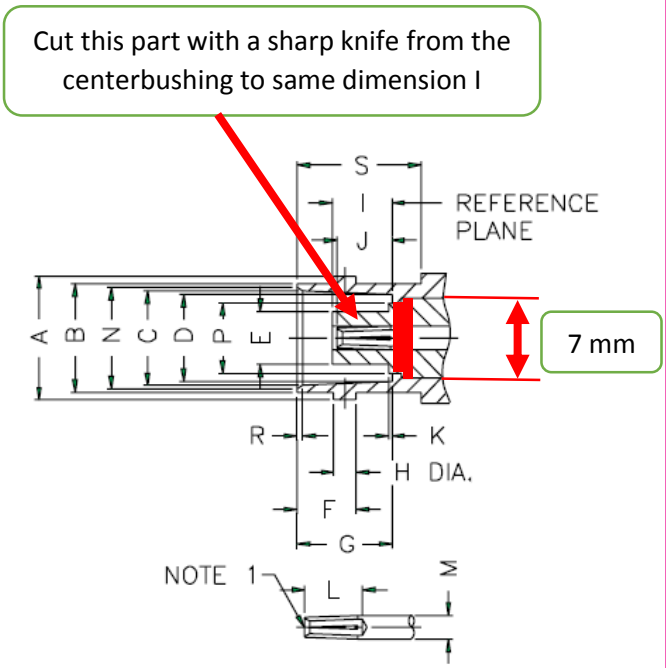
	Plug		Jack	
	min.	max.	min.	max.
A	4.83/.190	–	–	4.72/.186
B	5.33/.210	5.84/.230	4.72/.186	5.23/.206
C	5.28/.208	5.79/.228	4.78/.188	5.28/.208
D	2.06/.081	2.21/.087	2.06/.081	2.21/.087
E	9.78/.385	9.91/.390	9.60/.378	9.70/.382
F	1.98/.078	–	4.95/.195	–
G	1.32/.052	1.37/.054	8.31/.327	8.51/.335
H	0.08/.003	–	8.10/.319	8.15/.321
I	–	–	10.52/.414	–

Interface dimensions conformable to the standards:

International:	IEC 61169-8
Europe:	CECC 22120
USA:	MIL-C-39012, BNC Interface MIL-STD-348A/301
Great Britain:	BS 9210 N 004

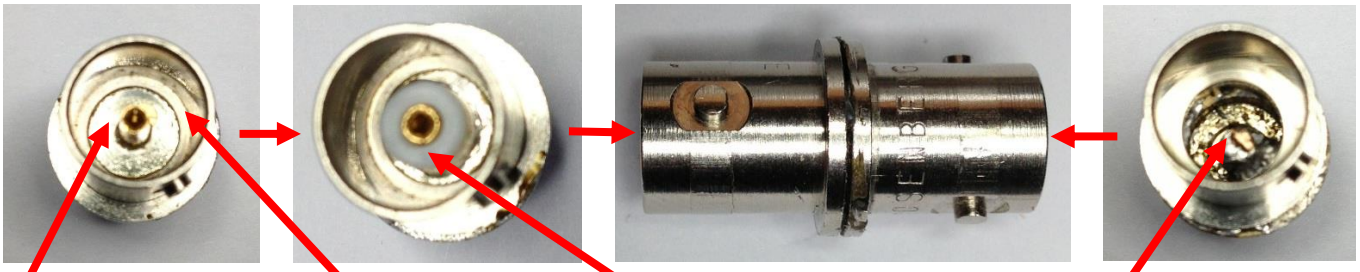
This type is a good candidate. I used a similar type with same dimension from another supplier.
WARNING BEWARE OF USING 75 OHM CONNECTORS AND THRU ADAPTORS USE ONLY 50 OHM

Design of the "0ps" Short Reference Standard:



Letter	Millimeters [Inches]	
	Minimum	Maximum
A	10.97 [.432]	11.07 [.436]
B	9.60 [.378]	9.70 [.382]
C	8.31 [.327]	8.46 [.333]
D	8.10 [.319]	8.15 [.321]
E	–	4.72 [.186]
F	5.18 [.204]	5.28 [.208]
G	8.31 [.327]	8.51 [.335]
H	1.91 [.075]	2.06 [.081]
I	4.78 [.188]	5.28 [.208]
J	4.72 [.186]	5.23 [.206]
K	–	0.15 [.006]
L	4.95 [.195]	–
M	2.06 [.081]	2.21 [.087]
N	8.79 [.346]	9.04 [.356]
P	–	6.50 [.256]
R	0.38 [.015]	0.76 [.030]
S	10.52 [.414]	–

Produce a small disk e.g. 2mm thick, as drawn with red above, and ensure the surface of diameter P 6.5mm (about 0.7mm deep) is flush with the reference plane. The disk is 7 mm in diameter to the right side and make it to a press fit. Drill a 2.2.mm hole in the center of the disk and solder the center pin to the disk from the read side and ensure the dimension J is maintained. You may mount the assembly on a male BNC connector to ensure center pin is placed in the center while soldering. If you start with drilling a 2 mm hole and mount the disk on a Dreml spindle, then you may grind it down to 7 mm against a square file and also produce the 6.5mm recess of depth 0.7mm. Then the hole later can be expanded to 2.2mm with a "rat tail". ***My shorting disk was not 100% in the calibration plane but 0.1mm recessed so that will be encountered for during calculation of the real "0ps" delay.***



Shorting disk flush with the calibration plane PTFE Bushing fitted Shorting Center pin soldered from this side

Perfect 0 ps Short soldered together to and "empty Open" which might be used as alternative Open Standard
The PTFE center bushing may need a small drop of 1 sec glue, but never put glue on the reference plane, outside the PTFE bushing or on the center pin. Just a sparse drop under the seat of the bushing prior to insertion.

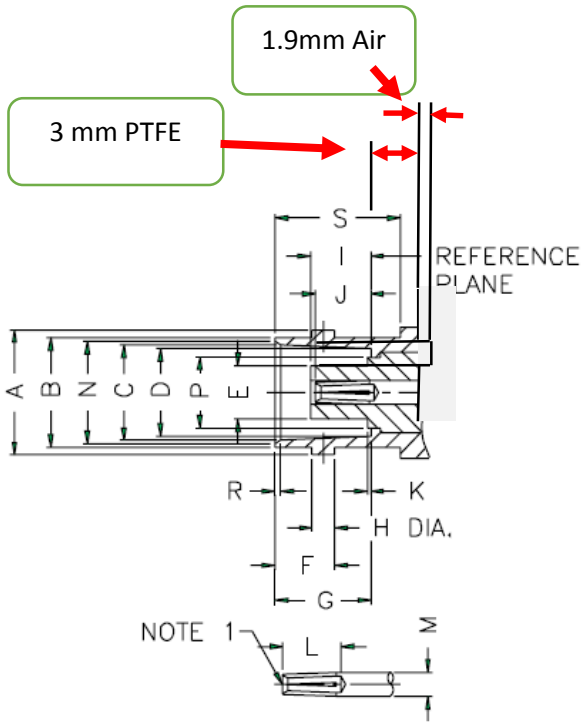
Production of the Load and Short

From a mechanical stability point of view, it is not advisable to produce a short Load Standard like the 0 ps Short Reference Standard, as the center pin cannot be secured, and the two 100ohm 0.1% SMD 0805 resistor will break when handling during calibration. That is why these two devices are made in full length and as no grinding of the metal parts are done, they will be of same electrical length. There will be a small additional fringe capacitance from the Load centerpin to take into account. Be very sparse with solder used on the center pin as that adds to the fringe capacitance, which will be in the region of 50 fF (more on that during T-Check tuning). Remember to grind the center pin down to the same plane as the inside side of shorting disk, on the Reference Short, which means to the end surface of the threated metal part.

This page is only part of the document as an illustration how to perform calculation of delays.
It will be needed to find the true delay of the "0ps" Reference Short

Production of the Open standard with calculated data based on mechanical dimension.
It is not the type of Open which will be the final but just shown for how to do it !!!!!!

Grind down to the flange, while a Male BNC Male connector fitted. The Centerpin is pushed forward during that process and pull I backwards until the dimension J is restored. Use a drop of 1 sec glue to secure from the back while sliding the center pin inwards and remove excess glue. Beware no to add glue to the slotted part of centerpin.



y

Letter	Millimeters [Inches]	
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S	10.52 [.414]	–

Calculation of the delay:

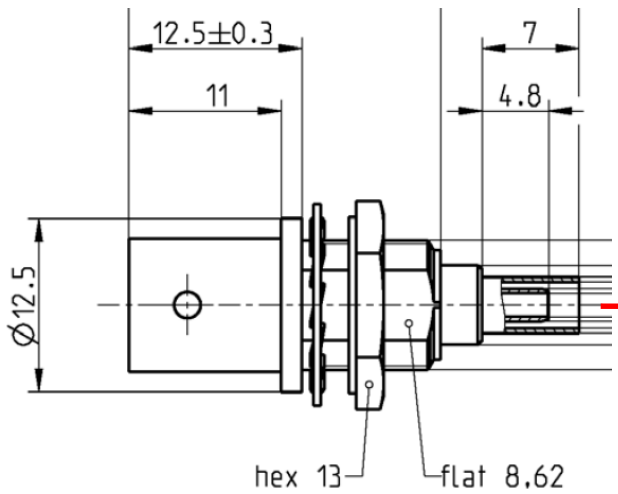
3 mm PTFE measured from the reference plane with velocity factor 0.69 provides 3.3333/0.69x3=14.5ps
1.9mm airline with velocity factor 1 provides 3.33333x1.9=6.3ps
Total delay 20.8pS which in the calibration setting must be entered as 2x20.8=41.6ps
Fringe capacitance will be about 25-75 fF.

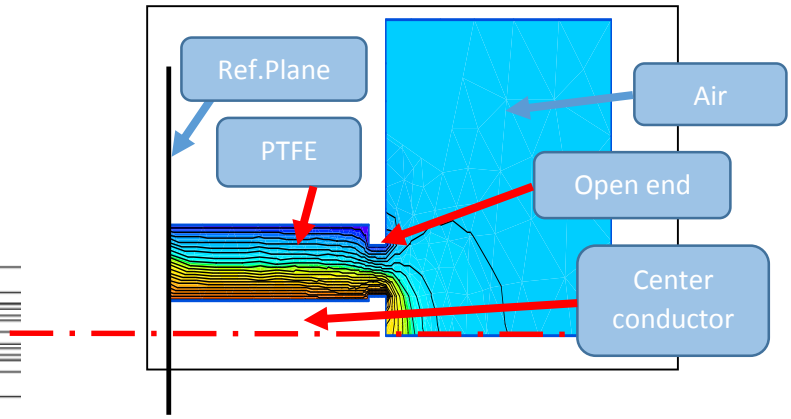
Calculation of the real delay for the "0ps" Reference Short

As the shorting disk is recessed 0.1mm then the delay is 3.33ps x 0.1= **0.33pS**

Design of the best type of open for the final kit

Using the same length for the open standard is a best choice as the Short Open and Load then is of identical length and of known value due to the determination of the delay bases on the 0 ps Reference Short and subsequent derived from the "long" Short as shall be seen later.
The only "snake in the paradise" the fringe capacitance at the end of the Open, but that can be calculated based on a physical model by the program QuickField, of which there is a student version capable of doing that with high enough accuracy (said to be less 5%).





I have also measured the accurate distance between the reference plane and the end of the adaptor with my CNC machine to be 12.965mm. Thus I can simulate the delay and compare to the measurements performed later on.
Only half of the BNC conductor drawn in the simulator as it is circular around the centerline.
The capacity calculated with and without the air (of epsilon 1) at the end of the Open standard in the simulation. The fringe capacitance is then the difference between these two values and found to be **46.1 fF** equal to 2.3ps delay.

The conditions for these calculations was a center conductor diameter of 2.15 mm and the internal diameter of the BNC adaptor 7.1mm and the diameter of the reduced hole, where the center conductor extrudes the metallic end was 5.7mm. Some BNC conductors has a smaller diameter and for 5 mm the fringe capacitance is only increased to **46.5fF**

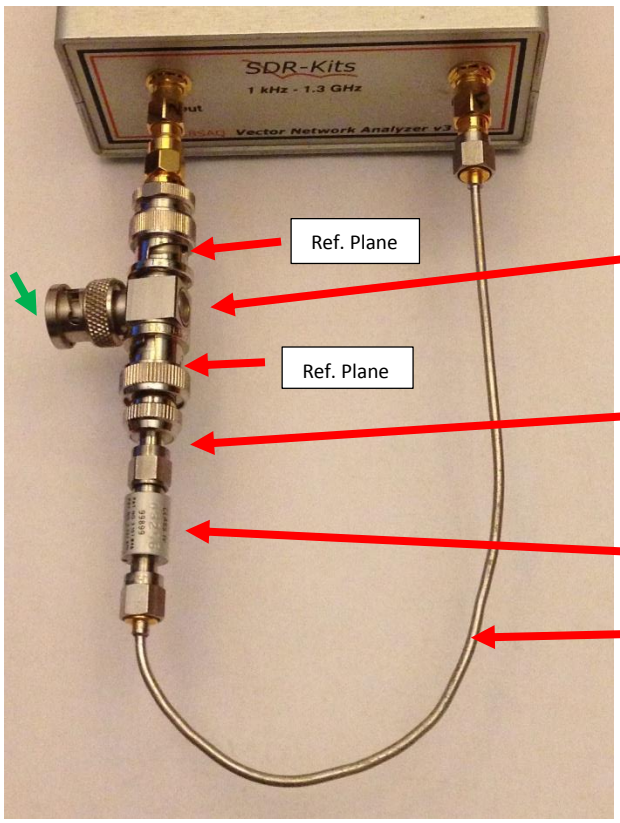
The delay of the adaptor (length 12.965mm) calculated to be 68.32ps for Epsilon 2.05 and 66.68ps for Epsilon 2.0 and for 5mm opening 70.04ps for Epsilon 2.05 and 68.33ps for Epsilon 2.0 .
(For the simulation was used a length of 13mm for practical reasons and found to be 68.5ps for Epsilon 2.05 and 66.86ps for Epsilon 2.0 and for 5mm opening 70.23ps for Epsilon 2.05 and 68.51ps for Epsilon 2.0 . As the delay is a linear function of length then it is easy to find YOUR DELAY). The Epsilon figure is equivalent with velocity factor and is either 0.69 or 0.7 pending where you find the information. Huber Suhner informs Epsilon is 2.05 and that is why I have underlines usage of 2.05.

Also for the Load was simulated the **applied solder** to the two 100 ohm SMD resistors of **height 0.8** mm and that amounted to additional **3fF** equal to a delay of 0.15ps so a very little contribution. Also the internal capacitance of the two SMD resistor could be considered added, but these are also very small values.

Pictures of the final BNC Female Open Standard of identical length to the Short and Load as seen to the right.

Fringe capacitance is **46.1 fF** or an additional delay of 2.3ps. For the Load the fringe capacitance is 3fF larger and thus incl. the capacity for the two SMD resistor a plain **50 fF** or 2.5ps delay is the end result.
Only remaining activity is to measure the common delay for the three standard compared to the simulation and perform a verification of the calibration quality using these the standards.

The Test Setup



The VNWA when used in Bridge mode seen on the picture.
A T-adaptor inserted in the signal path from TX to RX and the Female Calibration Standards fitted at the Green arrow during measurements

SMA to BNC adaptors used, which does not impact on measurement is calibration routines outbalances any imperfection.

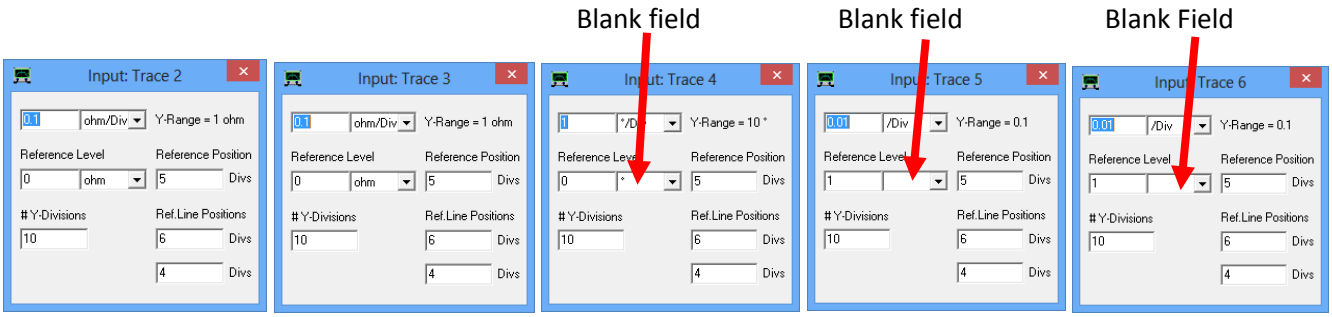
The 10 dB SMA inline attenuator used according to explanation already given.

A semi rigid cable used from the attenuator to RX port ensuring highest possible phase stability.
A normal test cable might induce small phase shifts when flexed.

ACTIONS TO PERFORM FOR THE VNWA SETTINGS

Some preparation is needed setting up the VNWA prior to entering of tentative calibration setting for our new female BNC Calibration Standard can be done:

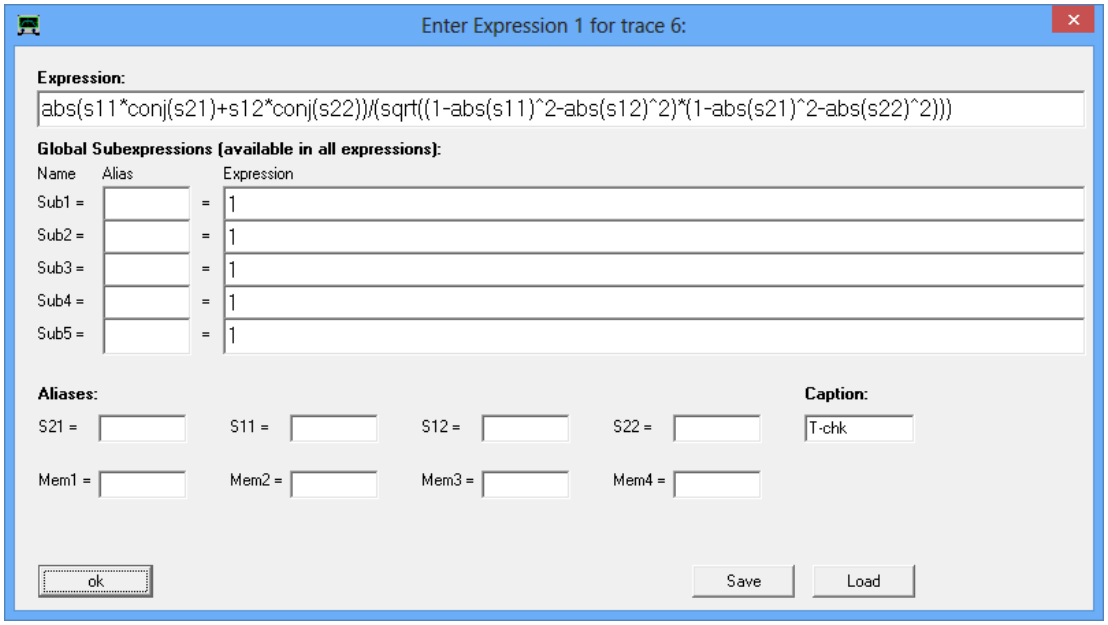
- Decide on Frequency span for which you want the calibration to cover. The VNWA are primarily designed for up to 500-600MHz and above to 1300MHz the performance is degrading but quite useable and we will in this tutorial cover 1 to 1300MHz.
- Active traces and set scale resolutions as described below where Trace1 is S11 Smith
- Set the time per point to min. 30 ms for optimum accuracy and thus the number of point will be 1000 for a sweep time of 30 sec. The number of points allows use of smoothing over 5-10 samples to reduce noise.
- As we are going to use a calibration method called Arbitrary Calibration, which might be new to most readers, it is recommended to study the help file on the subject, else just watch the next steps hopefully adequately explained.



Trace2 S11 Imag Z Trace3 S11 RealZ Trace4 S21 Phase Trace5 MEM2(lin magn.) Trace6 Cust. Trace (lin magn.)

For the custom Trace you need to enter the T-Check Expression which you just copy and paste into the **Expression: field** and also change the **Caption: field** to T-Chk, so you can see what the Custom Trace is all about.

abs(s11*conj(s21)+s12*conj(s22))/(sqrt((1-abs(s11)^2-abs(s12)^2)*(1-abs(s21)^2-abs(s22)^2)))



At first we will do a little stunt

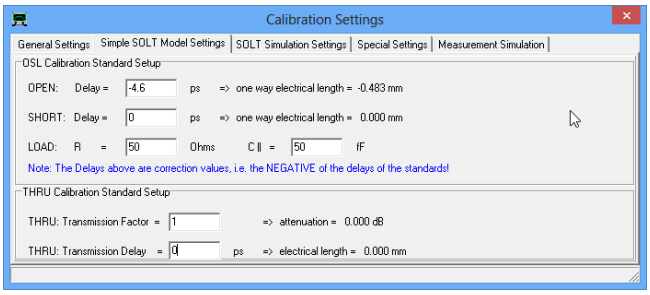
At first we will use the VNWA in a special operation mode called External Bridge and insert a T-Adaptor in the signal path between TX and RX Port particular and use the “third method” for measuring impedances particular accurate for measuring low impedances. The setup is shown on previous page, and a 10dB inline attenuator inserted in the signal path to linearize toward 50 ohm of the input impedance of the RX port. (See the remark in the final summary where comments made that you may not need this)

<u>The stunt is the following:</u>

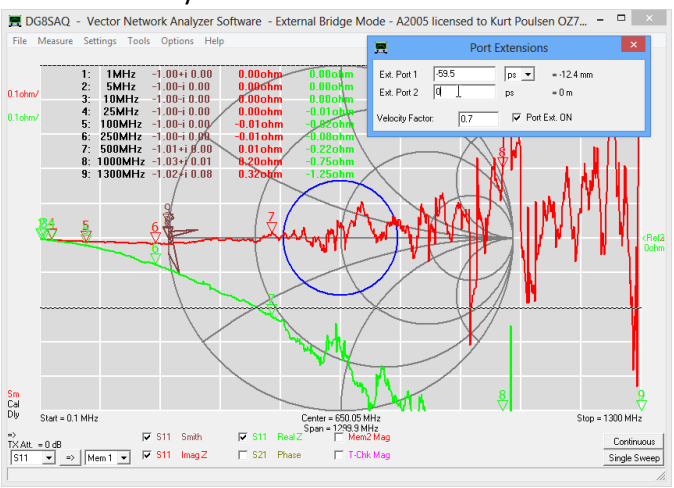
We will calibrate to the end of the BNC Adaptors, constituting the real Open Short and Load standards, and the only use the fringe components as calibration standards values. Then we afterwards measure the “0ps” Short Reference Standard. By this method we can compensate for the different length of “mine and yours” BC female adaptors provided you produce a “0ps” Short Reference Standard.

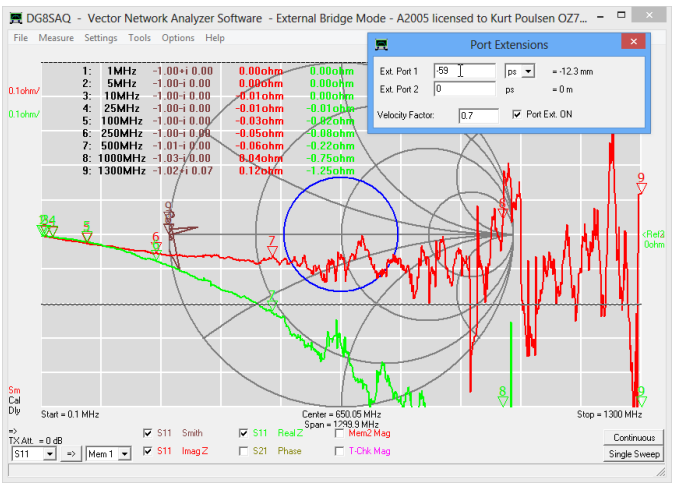
A footnote: Actually is in the real practical world, it is not so essential that you calibrate exactly to the reference plane of the connector type used. What important is that you calibrate the Open, Short and Load and Thru to the same plane, whether it is a slightly away from the connector reference plane or not. Using this “stunt” calibration can always be used and then a Ext. Port 1 delay used to move the calibration plane to the connector mechanical reference plane.

Calibration setting is then as follows 2x -2.3ps for Open, 0ps for Short and 50ohm/Cp=50fF for Load, saved under the calibration kit name “BNC Female Stunt”. Thru is not relevant as the External Bridge VNWA Operation mode only is for reflection measurements.



After a careful calibration in the order Load, Open and Short follow by immediate measurement of the “0pS” Short Reference Standard, to minimize drift of the calibration, we observe that the ImagZ is not a flat trace after entering a Ext. Port 1 delay.

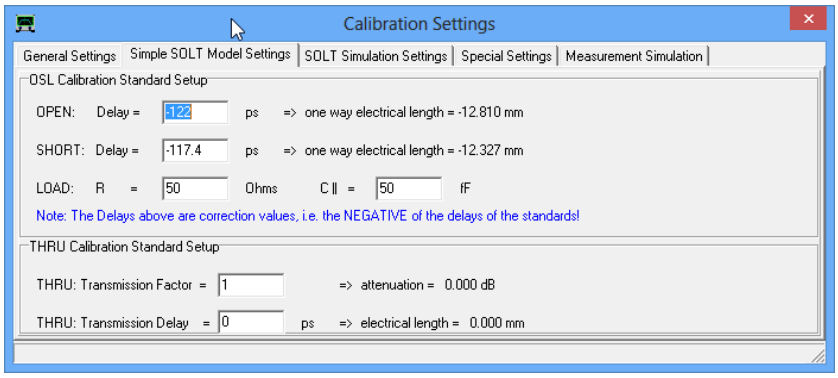




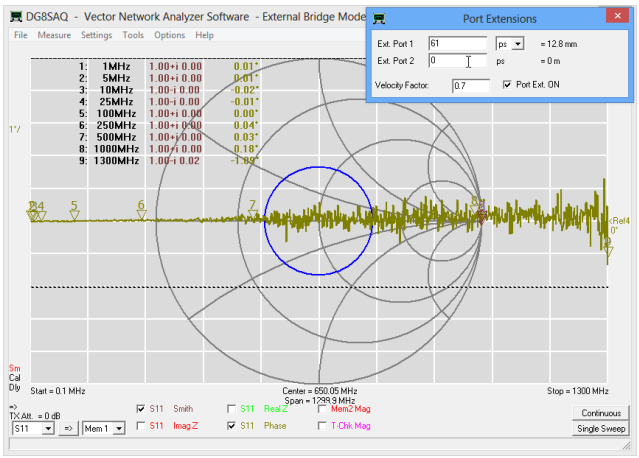
Is 59.5 ps or 59.0 ps the right value ? Actually if we use the VNWA below 500MHz we would use 59.5ps but the right value is merely 59.5ps as the inductance and resistive part of the “0ps” Short Reference standard is not taking into account. In a transmission line the velocity factor is lower at low frequency (and then a shorter delay) and start to approach a constant value at some 200MHz (Credit to Dan AC6LA for illustrating that effect in the spreadsheet TLDetails) so for the time being let is stick to 59ps as the value where the plot is asymptotic flattening out above 500MHz. This effect can be eliminated by arbitrary calibration setting subject for another report later on. We also have to subtract 0.33ps as the “0ps” Short Reference Standard was not 0ps exactly so **net result chosen 58.7pS**

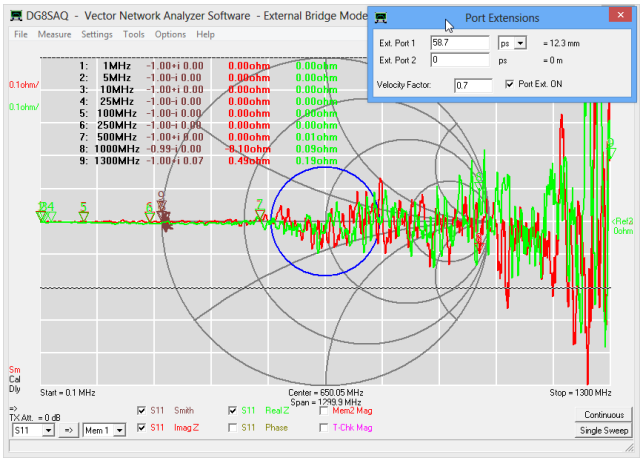
Beyond that removing/fitting the “0ps” Short Reference standard changes the measurement by + - 0.5pS easily. That is the penalty by using the BNC connectors.

The real calibration settings for out Female Calibration Standards

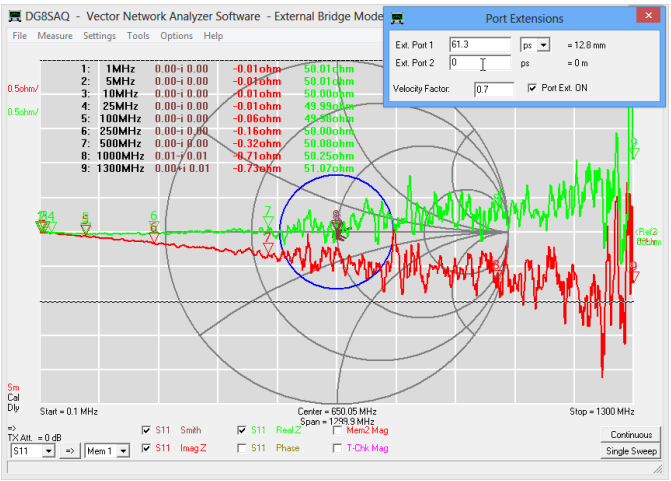


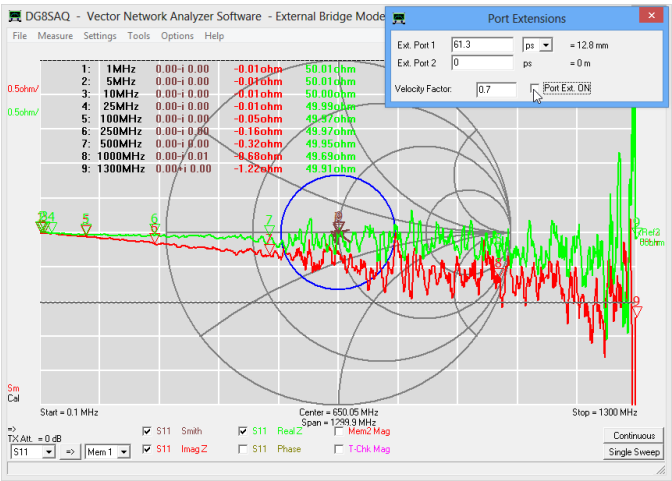
In above is entered 117.4ps for Short, 122ps for Open and for the Resistance 50ohm with fringe capacitance + the small addition for solder in total 50fF.





After calibration a measurement of the Open (left above) seem OK with the Ext. port 1 delay set to 61ps as was entered in the calibration settings so the open is OK. To the right is the Short measured and the effect earlier mentioned for the inductance and resistance is a appearing above 500MHz but it is only due to this effect mentioned Let us see how the load behaves.

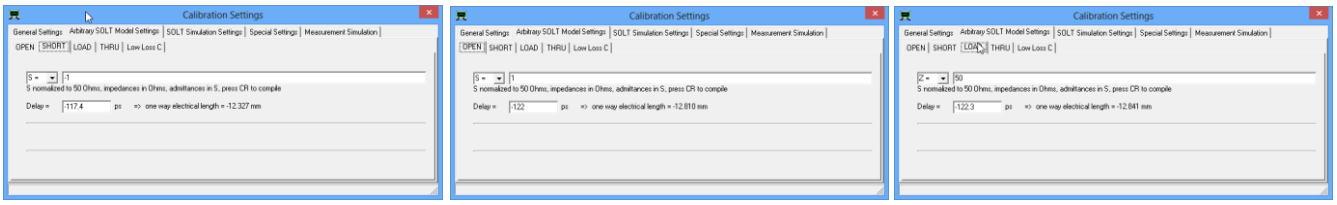




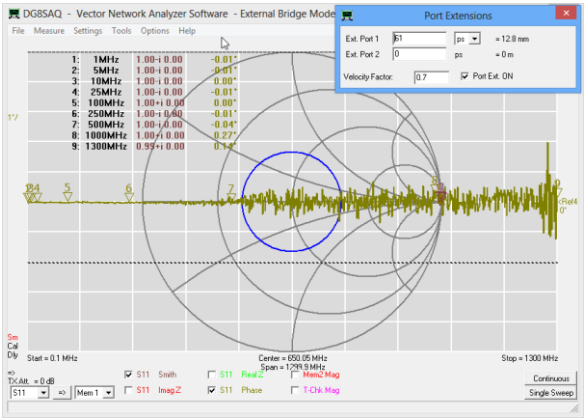
Above left is the load measured and Ext port1 delay set to the calculated delay incl. fringe capacitance and the right image is for no Ext.port1 delay. In both cases the scale changed to 0.5ohm/div and both the resistive and reactive part is absolutely not as expected. This is simply the problem which happens when a discrete load incl. fringe capacitance is placed at the end of a transmission line as impedance transformations take place. Just think of the situation where and open quarter wave long transmission line is none terminated then at input it is a short and vice versa. Even for so short a transmission line as our Load Standard it is a problem. However in the VNWA calibration

Settings we can compensate to some degree using arbitrary calibration settings and enter a delay. However this is under the assumption of an ideal transmission line but let us see how far we get by this method.

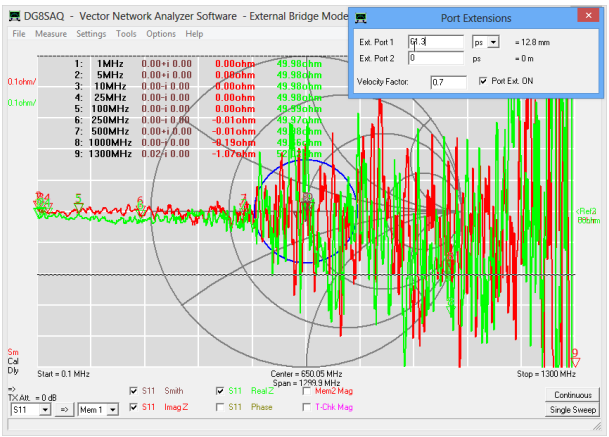
Arbitrary calibration settings

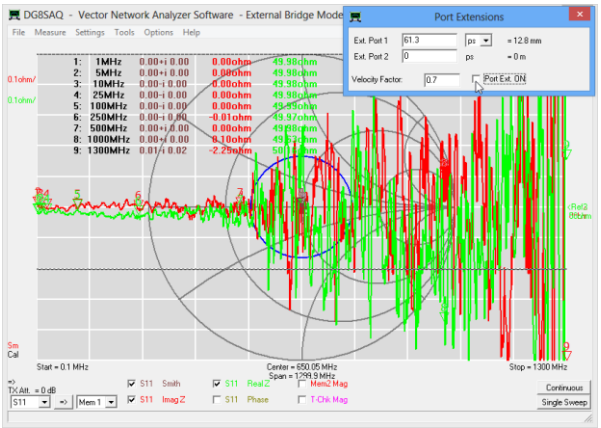


In above is entered 117.4ps for Short, 122ps for Open and 122.3ps and the Resistance 50ohm. The fringe capacitance like as for Open + the small addition for solder is included in the delay.

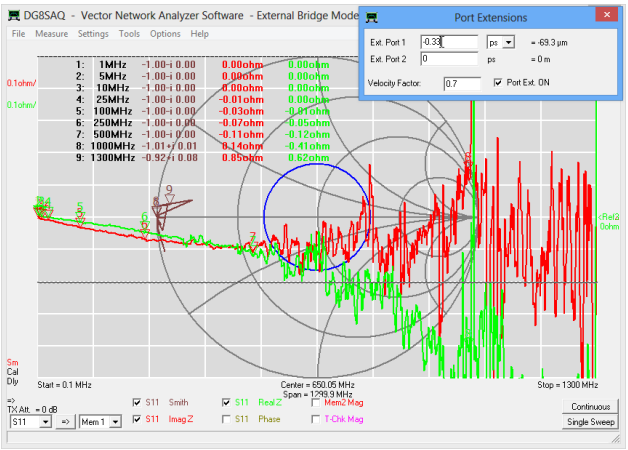


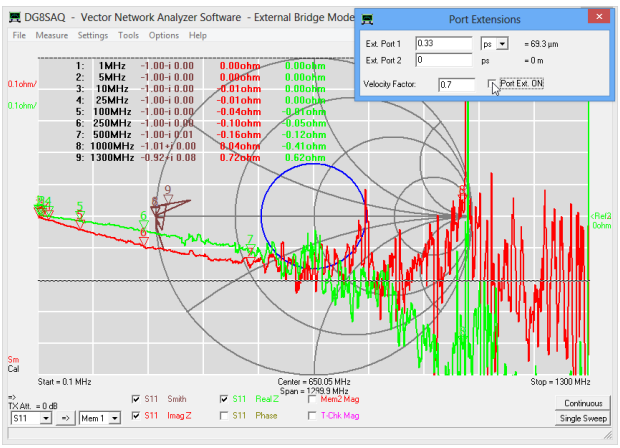
For Short to the left the trace are improved and to the right the open is as before OK





For the load a significant improvement and even without Ext. Port delay the impedance is more as expected



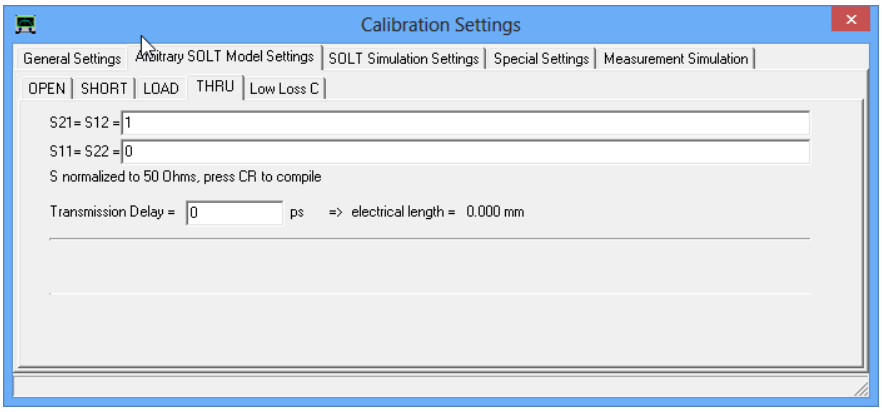


Measurements with the calculated delay 0.33ps for left image and without for the right image the ”0ps” Short Reference Standard indicates we are very close to optimum data for our BNC Female calibration standards.

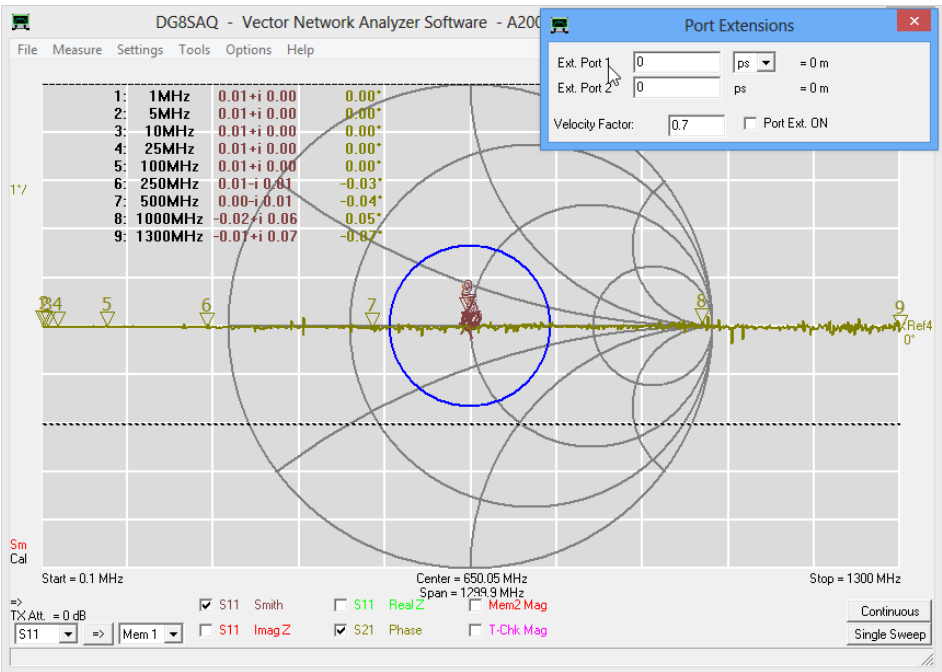
A preliminary summary:
Using arbitrary calibration is a must when the load resistor consist of a discrete resistors at the end of a coaxial structure. Determination of the delay is so far within 0.5ps after best estimate and probably difficult to do much better, unless full correction made for inductance, resistance and losses as mentioned before due to the effect of frequency dependent velocity factor below some 200MHz. The simulated delays was 68.32ps for Epsilon 2.05 and 66.68ps for Epsilon 2.0. The measured delay of 58.7ps and the difference can be explained by the BNC adaptor construction as below the reference plane there is a section of about 1mm not filled with PTFE and the BNC mechanics has many transition not included in the modelling.

Next step is performing a test with the VNWA in Standard Operation mode and performing a T-Check.

However first we need to determine the delay of the Thru adaptor and the procedure to use is to calibration S21 Thru and Thru match with the Thru adaptor fitted and in calibration setting set the delay of the Thru adaptor to 0ps. Then we do a complete SOLT calibration, followed by a T-Check with the T-Check adaptor of same length fitted. The T-Check result will of course look awful but then we apply the smart function "Enable Realtime recalibration" in the calibration settings and adjust the Thru adaptor delay until the T-Check trace is lifted to it's "top value" or as close to the horizontal o % line, and shows the best result. Then we have found the delay of the Thru adaptor under the condition the Thru adaptor and the T-Check adaptor is identical delay/length from the beginning.
Let see how that works out.

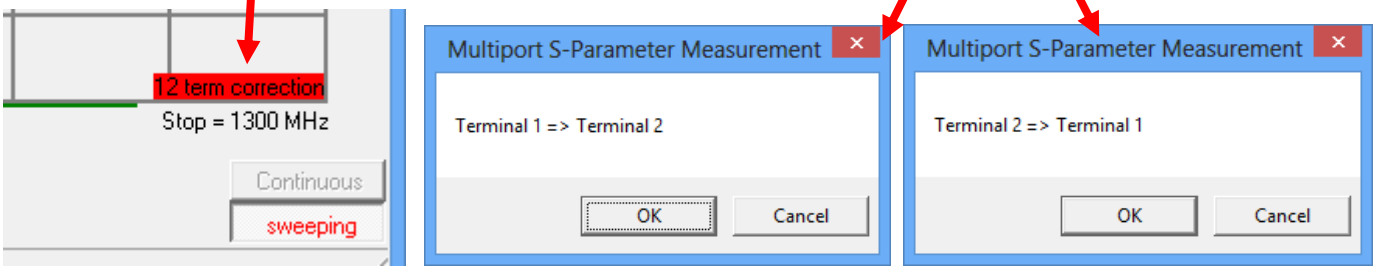


Setting of the 0ps delay for the Thru adaptor



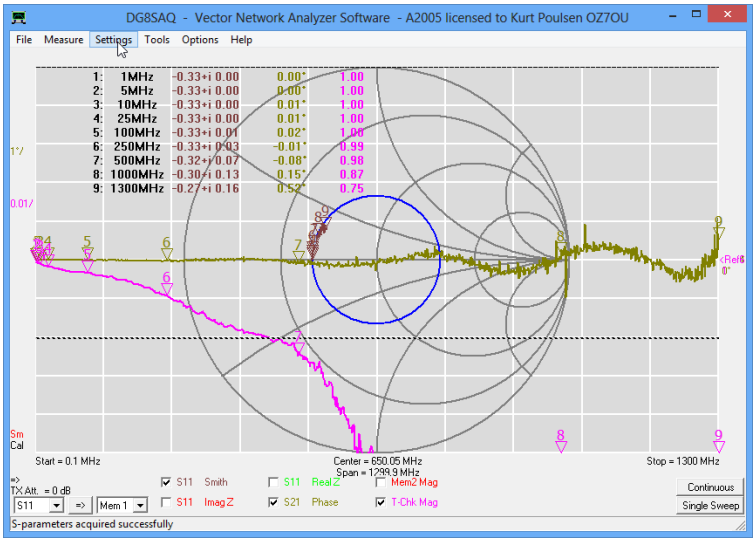
A S21 Sweep made and as calibrated the 0 phase shift for S21 in the calibration plane as we have decided so.

Now we fit the T-Check adaptor instead of the Thru adaptor and perform a T-Check and we do another stunt by pressing the **F2 key,** as we then perform a full 10/12 Term calibration, with only a one port calibration (5/6 term), as indicated by a small red Icon in the lower right corner, as seen below. Follow the instruction on the screen and when you are asked for running reverse direction (Terminal 2 => Terminal 1) then the T-Check adaptor must actually be reversed, but if it is 100 symmetrical, you may as well cheat on that point.



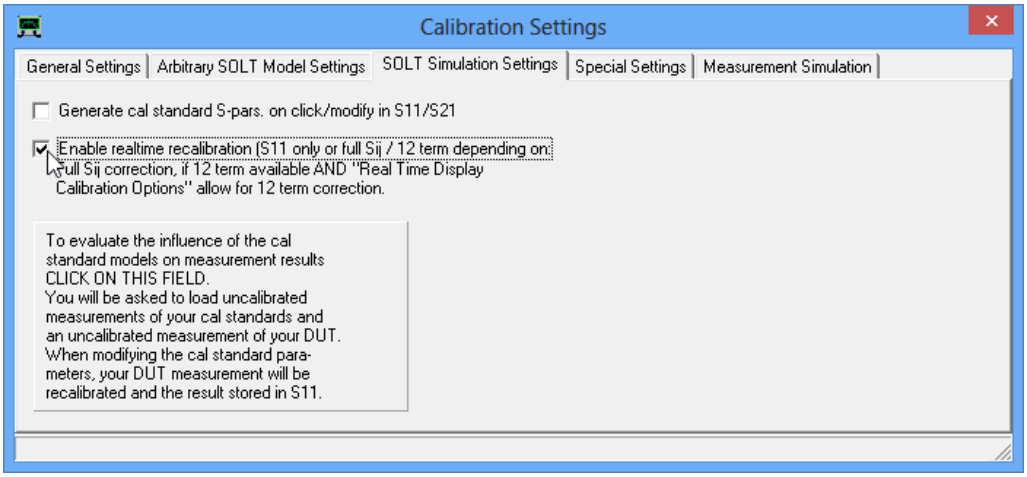
Remark! To run a full 10/12 term error correction you normally need to use a Test Set, which is reversing the measurement direction for the DUT automatic or in the Setup chose a Test Set with manual switch operation and revert the DUT by hand. It also requires to perform a full SOLT calibration in both forward and reverse direction. The F2 method is a superb method for doing a full two port measurement without having a Test Set and only perform a one port calibration.

You may measure manually first S11 and S21 and the reverse the measurement direction manually and then measure S2 and S12 (without enabling a Test Set in the Setup) but these measurements are only 5/6 Term error corrected measurements and thus provide poorer T-Check results.

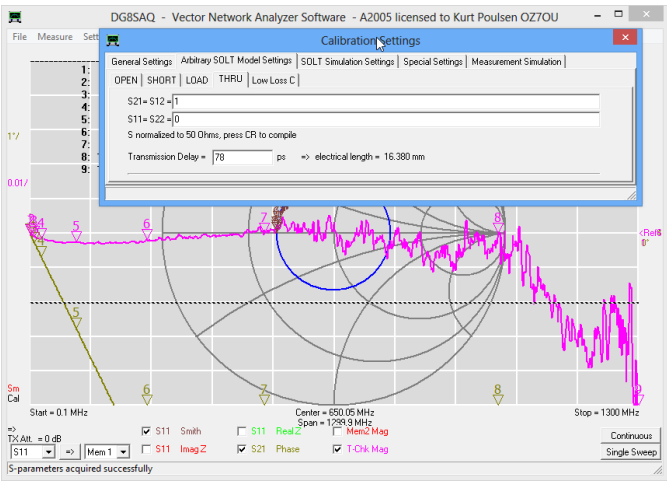


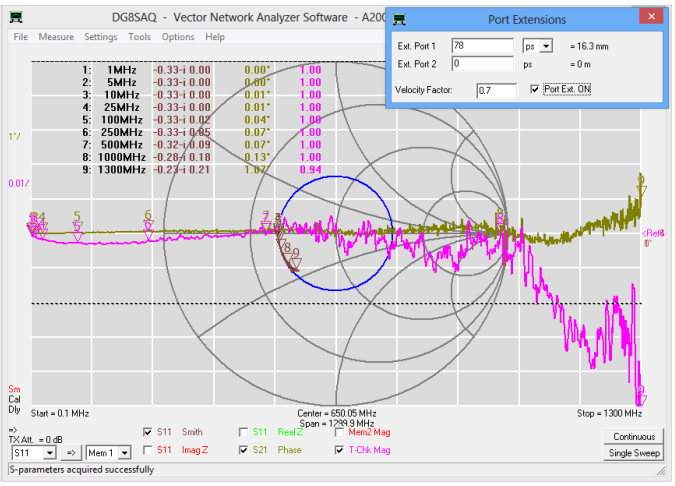
The T-Check trace (magenta) is off course “faulty” as the condition for a T-heck not met.

Now open the calibration setting and tick mark the “Enable realtime recalibration” which allow us to tune the calibration settings without running a new calibration and a new sweep.



In calibration setting select the THRU tab and change the Transmission delay (left image) until the T-Check Trace is elevated and start to “back” to previous values. You will see that for frequencies up to 500MHz the result is far better 0.01 as equals 1%. If you enable Ext. Port 1 delay (right image) and enter the found delay for the Thru adaptor, here 78ps, then the S21 phase is back to 0 degree up to more than 500MHz, as we now is measuring in the SOL calibration plane again now also the S21 calibration plane.

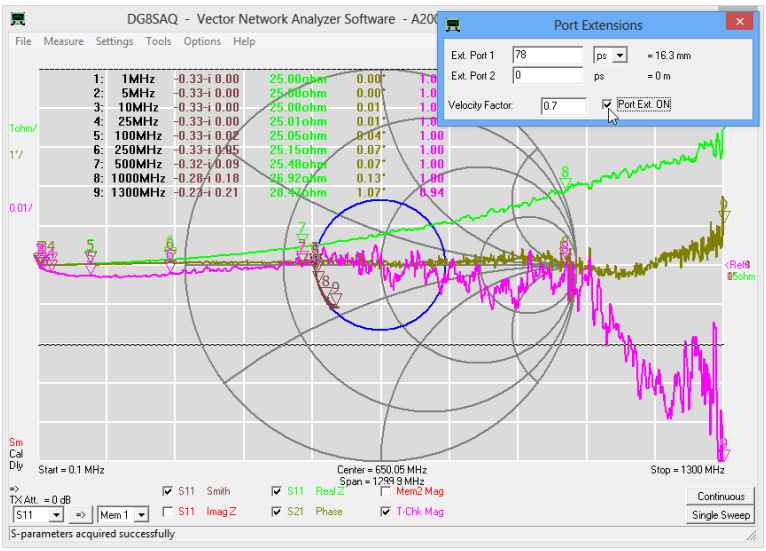




By changing the Real Z reference to 25 ohm we see in below image the resistance is 25 ohm, which is constituted by the 50 ohm for the T-adaptor in “parallel” with the input impedance of the RX port through the 10dB inline attenuator and it is quite ok that it is not straight 25ohm, as we now measure these two impedances through a 50 ohm transmission line, and that causes impedance transformations as described earlier. Just another demonstration of that effect.

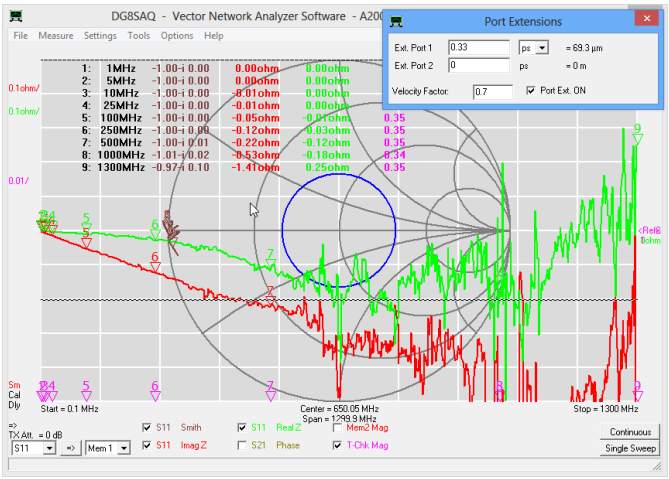
We have now finalized the characterization of our new complete BNC female SOLT calibration kit and only need to check if there is any fine tuning possible.

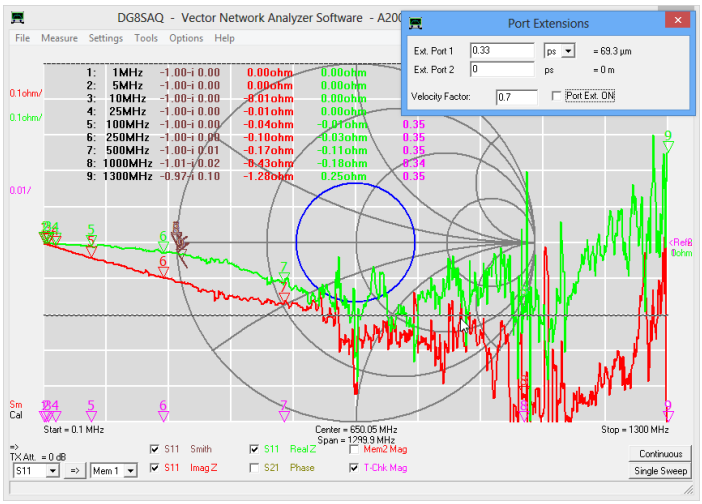
As all our BNC adaptor delays was determined in the External Bridge mode we may see it the standard VNWA mode measures the “0ps” Short with same results. However we ought to save the calibration settings at this points in time under a descriptive name.



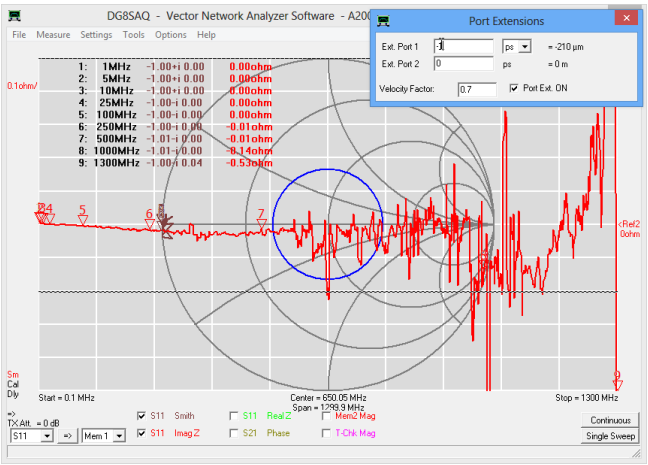
Test of the performance I normal VNWA mode with a fresh calibration.

Below is shown the measurement of the “0ps” Short Reference Standard. Same result as before as 0ps Ext. Port delay



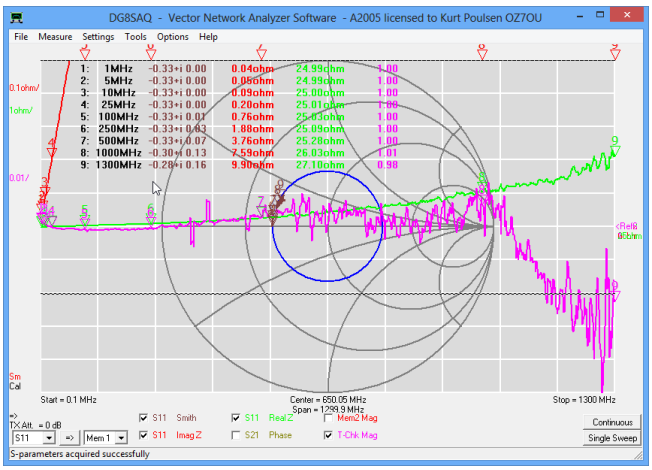


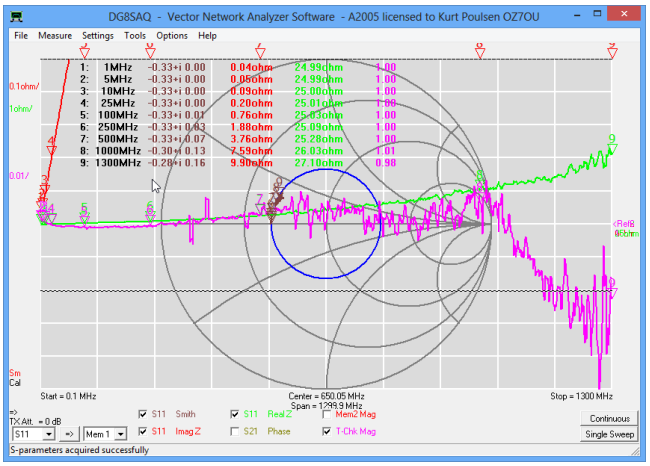
And below is seen that -1ps brings it to the ImagZ =0 line



If you recall earlier I debated if the delay should be 59.0 or 59.5ps. Hm… maybe 59.0 was better ☺

Below to totally independent T-Check made with new calibrations





My Goal obtained for a BNC Calibration Kit with a T- Check less than 1% to 1GHz and below 3% to 1.3GHz. It is probably not realistic to aim for better performance without using measurements against a professional standard and more sophisticated arbitrary calibration methods.

Just the mechanical removal and refitting of the BNC Calibration Standard yield some + - 1ps variation, so that alone implies this simple production method is optimal.

Reproduction without having a “0 ps” Short Reference Standard is possible some careful thinking and corrections of the delays in this report based on the mechanical dimension of you adaptors. The fringe capacitance for the Open is given for 5 and 5.7mm end radius. The fringe capacitance for the Load is same as for the open with the very small additional delay caused by solder. So if you produce a T-Check adaptor you can find the Delay of the Thru adaptor else difficult to determine as it in most cases has sections of airlines without (PTFE) and invisible to see.

A work Plan for a ”no fuzz” cloning of my design.

1. Purchase the 5 adaptor from a well know source. Check their homepage for mechanical dimension drawings- Many manufacturers does not provide accurate dimension for the total length and inside diameters, so you need to have a good caliper.
2. Manufacture the Open Short and Load as per my description. Also produce the T-Check adaptor as described. It is not recommended to use the produced Load together with a T adaptor, of appropriate gender, as you then cannot determine the Delay of the Thru adaptor.
3. If the total length of your adaptors differ from mine, which is 21.4mm (measured on the Open Standard) then you quite accurate can calculate the shorter/longer delay based on 4.83ps per mm.
4. Use for fringe capacitance for the open a delay of 2.5ps
5. For the Load use same fringe capacitance as the Open and add further 0.2ps
6. Do a T-Check Test after a fresh calibration and test also up to 1.3GHz. You may fine tune then with the “real time recalibration feature” but adjust the open, short and load with same about. Most likely the Thru adaptor delay will be the one which might get the T-Check to fit exactly along the horizontal line and focus main up to 500-600MHz. If you are getting below 1% up to 500 MHz and below 2-3% at 1 GHz you have an excellent BNC Female Calibration Kit.

A note for the N2PK owners. Remember the VNWA software is having a VNWA mode so quite handy for the tasks here described.
A male calibration kit can be designed along the same principles as outlined in this report.

Have fun.

Januar 2. 2014 Kurt Poulsen Text bug page 1 corrected Jan. 4 2014